

Website Link with Informative Titles for each Page

<http://classwork.engr.oregonstate.edu:8000/Avidex>

Avian Compendium

- A list of all the birds held within the Avidex servers that users can find and track on the site

Rarities

- A list of the rarity types for all the birds

Birders

- A list of all the Birders that have an account

Nests

- A list of all the possible nest types (A cross product of the location and type of nest)

Rewards

- A list of all the rewards the birds can get

Sightings

- A list of all the sightings that a user has made

BirdersRewards

- An intersection table holding each Birders' Rewards

BirdsList

- A list of the total sightings of each specific bird a user has seen

BirdsNests

- A list detailing each bird's type of nest

Peer and TA Review Feedback on Project Draft 1

Daniella Norris

Hey friends,

I love the birding collection app idea! I know my friend that enjoys birding always struggles to keep track of sightings without having a digital interface to track. I also like that you included research to back up the intentional nature of your DB design. Well done!

Edited this post to match the required format!

- Does the overview describe what problem is to be solved by a website with DB back end? If yes, summarize. If not, what changes would better support describing the problem to be solved?
 - Yes. The overview clearly explains that the application is intended to help birding enthusiasts and newcomers record and track bird sightings in a fun, gamified way. It also explains how the database will support collecting bird species and storing sightings, giving readers a clear understanding of the project's purpose.

- Does the overview list specific facts? If yes, summarize what the facts illustrate about the proposed DB solution. If not, what facts would better support illustrating the scope and scale of the proposed DB solution?
 - Yes. The overview includes details about the application's target audience and the gamified "Pokédex-like" experience, which helps explain the intended functionality of the database. It would be even stronger if it included estimated numbers of users, bird species, or expected sightings to better communicate the anticipated scale of the system.
- Are at least four entities described, and does each one represent a single idea to be stored as a list? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?
 - Yes. The proposal includes several well-defined entities, including Birds, Rarities, BirdsNests, Nests, and Sightings, with each representing a single concept appropriate for a relational database.
- Does the outline of entity details describe the purpose of each, list attribute datatypes and constraints, and describe relationships between entities? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?
 - Mostly. The entities include appropriate attributes, data types, constraints, and relationship descriptions. One suggestion would be to describe relationships using the Entity.attribute format (for example, Birds.birdID has a 1:M relationship with Sightings.birdID). This makes it much easier to identify which primary and foreign keys are related and reduces ambiguity when implementing the schema.
- Are 1:M relationships correctly formulated? Is there at least one M:M relationship? Does the ERD present a logical view of the database? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?
 - Yes. The ERD presents a logical design with several one-to-many relationships and a many-to-many relationship between Birds and Nests resolved through the BirdsNests junction table. One small issue I noticed is that the Nests entity appears to be missing its primary key in the ER diagram, even though it is included in the database outline. You may want to update the ERD so it matches the written schema.
- Is there consistency in a) naming between overview and entity/attributes b) entities plural, attributes singular c) use of capitalization for naming? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?
 - Yes. Overall, the naming conventions are consistent throughout the proposal. Entities are plural, attributes are singular, and capitalization is applied consistently. The only recommendation is to ensure the entity names are consistent between the outline and relationship descriptions (for example, using either BirdsNests or BirdNest, but not both).

- As a reviewer, clearly describe to what extent your feedback to the team was original (e.g. "all my work") or non-original (e.g. used AI tools per Code Citation Tips).
 - All work on this post is my own.

Alec Moschetti

Hi Ayush and Joseph!

Wonderful idea for a database project, I love the idea of a Pokedex for real-life birding.

I appreciate the formatting of your project proposal that you did, it gives it a nice and formal touch. I think what specific problem this database is solving is answered but it's a little implied. I.e., you mention that it will encourage users to go out into the world and look for birds to earn incremental rewards. This is a solution but to which stated problem? Perhaps you want to write about how bird watching newcomers is on the decline or another problem statement that this solves specifically. Additionally, you could use that problem statement to include some numbers or statistics about birding or how this will increase birding and thus bird populations since you'll have more interest in protecting them. Obviously it doesn't have to be that but just an idea or example of what could be put here for the overview.

You've got 5 entities here (Birds, Rarities, BirdsNests, Nests, Sightings) which does meet the project requirements.

Birds is your most important and thus highest attributed entity. I would suggest that you maybe drop the Genus attribute since Species captures that information or at the very least make it not a unique constraint since many different species of birds will share the same genus. Perhaps also Wingspan and Size attributes should be decimal place numbers instead of integers since rarely do measurements fall exclusively on an integer. I also am a little unsure about the Photographs attribute that it needs to be not NULL and that it is a varchar. In the next weeks lectures in the course, Professor Curry goes over what data type to use for binary objects (like photos), and suggests that a blob should be used for that. Lastly for the Birds attributes, I noticed nestTypeID as a foreign key as an attribute for Birds but don't see where that is defined in your other entities, I'm guessing it was meant to say nestID instead for the Nests entity.

For the Sightings entity, seenOrHeard as a tinyint is perhaps a slightly confusing boolean value. Does 1 mean it was seen and heard, or just seen, or just heard? Maybe you will have a better inclination but from this view I think it could be made more clear.

Would you possibly want a Users or Birders entity? If trying to match a personal Pokedex for birders to track their sightings in addition to the encyclopedic nature of the bird Pokedex, it might help to have a Users entity. There could be an interesting M:M join there between users and sightings as well or other insights you could pull out. Similarly, would it make sense to have a Ranges or Regions entity instead of just the attribute in the Birds entity? Ranges surely only has a few categories so could be a nice category table and then a M:M between Birds and

Categories. Just a thought if you're wanting to add more content to your Avidex. Otherwise yes all your relationships are formulated correctly and you have at least one M:M relationship.

There are a few naming and capitalization inconsistencies in your database outline. For instance in Birds, every attribute mostly starts with a capital letter but some contain spaces where others are camelCase. Some also contain spaces like Mating Season. Should be an easy fix but yes BirdsNests and Nests entities have the same consistency issues.

A good first draft and a lot of good possibilities with this project idea, I'm excited to see your next versions!

All of my above feedback came from me and only me no outside sources were used.

Jackson Marcum

Hello!

I thought this was a really good idea for your project! I enjoyed that your overview explained what the app is supposed to do rather than just listing the features and it made this very easy to understand and follow. I thought the ER diagram matched so well and connected it all together perfectly! One thing i would suggest is adding a little more detail about how the user would actually interact with the app. Right now I understand what information is stored but it would help to explain how someone would search for birds or unlock achievements or keep track of their own collections. But overall i thought this is a very good draft and everything fit well together. I think adding the extra explanations would make it even better than it already is!

Joshua Williams

This is a fun idea about a database for a bird-watching app that keeps track of different species of birds and when a user has seen them. It is similar to Pokemon Go, and users will fill out spots in their Avidex when they complete bird sightings.

There are four entities: "birds", "nests", "sightings", and "rarities". There are a lot of good attributes for these entities, but I have a few recommendations:

- For "rarities", I would add a "rarity_level" attribute or something to give each rarity a name, rather than relying on the id. This will make the meaning clearer.
- In "birds", "photographs" maybe should be singular. Otherwise, it could be a 1:M relationship with a photographs entity. If it represents something else like a url with all the photos, it should be labelled "photographsUrl".
- "Mating Season" should not have a space in it.
- "seenOrHeard" is of type tinyint. I am not familiar with that, but I do not know if that is the best type. Maybe explain how that will be used. It may make sense to instead have two boolean attributes like "wasSeen" and "wasHeard".

At a high level the ERD looks good, and the relationships are constructed correctly and are where they need to be. Here are recommendations regarding the ERD:

- Maybe "rarities" should have an optional relationship to "birds". I am also not sure if "birds" should require "nests". All the relationships in the ERD are currently required, so I think it may be worthwhile to double check each to see if any should be optional.
- The attributes in the diagram do not align exactly with the outline. I am not sure how much detail the diagram should have, but the attribute names should match at least, such as "birdID" and "bird_id".

Great job coming up with a unique concept and designing the database. I enjoyed looking over your work, and I appreciate that you took the time to do the research to make this very detailed.

This review is all my work.

Actions Based on Feedback Step 1

- Changed the notation in the ERD diagram: capitalized the entity names and changed snake_case attributes in the diagram to match the camelCase in the outline.
- Changed photographs to photographUrl
- Changed mating Season to matingSeason
- Changed the overview to include more information about the scope of our project
- Changed the overview to specify a statement of the problem
- Created a Birders entity to function as a users list
- Changed wingspan and size to a varchar to better describe a range of size
- Added a PK to BirdsNests
- Changed nestTypeID to properly represent BirdsNests' PK
- Changed where BirdsNests is named BirdNest to properly show BirdsNests
-

Team Design Changes

- Added descriptions to the attributes
- Created a Rewards entity to keep track of the various rewards available
- Created a BirdsList entity to function as a list that tracks a Birder's list of Birds they've seen

Peer and TA Review Feedback on Project Draft 2

George Cross

Hello Group 28,

Below is my peer review to your project proposal and data definition. I have posted responses below to each of the review questions stated in the Project Step 2 review assignment.

- **Q: Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?**
 - A: Yes, However, your Project Schema diagram has a lot of overlapping lines along crow-feet relationships. I suggest adjusting your diagram to reduce the amount of overlap in these relationships.
- **Q: Is there consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?**
 - A: Yes, the naming convention of entities and attributes are consistent throughout the project proposal.
- **Q: Is the schema easy to read (e.g. diagram is clear and readable with relationship lines not crossed)?**
 - A: The Schema is not easy to read, due to excessive overlapping of crowfeet relations. I see opportunities to clean this up slightly, such as with "BirdsList" and "Rewards" entities.
- **Q: Are intersection tables properly formed (e.g. two FKs and facilitate a M:N relationship)?**
 - A: Yes, the intersection tables are properly formed.
- **Q: Does the sample data suggest any non-normalized issues, e.g. partial dependencies or transitive dependencies?**
 - A: The sample data exceeds the recommended limit of 10 or less entries per table. This makes verifying data integrity very burdensome for the reviewer. I was unable to verify all the data because it was excessive. I suggest reducing the sample data to 10 or less entries per table so as to meet the assignment description and make reviewing for integrity easier.
- **Q: Is the SQL file syntactically correct? This can be easily verified by using PhPMyAdmin and your CS 340 database (do not forget to take backup of your own database before you do this!)**
 - A: When I try to run the Project SQL file, If I don't have my DB titled as "Avidex" it will not run. Based on reviewing other groups SQL files, there is a way to create the tables without specifying the name of the DB to be stored. I suggest looking at updating the SQL file so this is not an issue.
Also, the sample data is excessive for this step in the proposal and should be 10 or less entries per table, instead of 20 entries provided in some tables (e.g. Nests).
- **Q: In the SQL, are the data types appropriate considering the description of the attribute in the database outline?**
 - A: Yes, the data types are appropriate for each attribute given it's description.
- **Q: In the SQL, are the primary and foreign keys correctly defined when compared to the Schema? Are appropriate CASCADE operations declared?**
 - A: Yes, Primary Keys and Foreign Keys look correct. As far as I can tell, there is no need for CASCADE operations.
- **Q: In the SQL, is all example data shown in the PDF INSERTED?**

- A: Yes, all example data is both in the DDL as well as the pdf file. However, this sample data is excessive for this step in the proposal and should be 3-5 entries per table, instead of almost 10 as was provided in this proposal.
- **Q: In the SQL, are relationship tables present when compared to the ERD/Schema?**
 - A: Yes, the relationship tables are present in the ERD and Schema.
- **Q: Is the SQL well-structured and commented (e.g. hand authored) or not (e.g. exported from MySQL)?**
 - A: Yes, the SQL file is structured neatly and executes/imports correctly.

Statement of Originality: All responses to the CS340 Project Step 2 review questions stated above are entirely my own. The question prompts were taken from Oregon State University's CS340 (CS_340_400_U2026) Project Step 2 Review assignment page on Canvas.

Work Cited: Oregon State University. (n.d.). CS 340 - Project Step 2 Review [Canvas assignment]. Canvas. Retrieved July 27, 2026, from https://canvas.oregonstate.edu/courses/2051721/assignments/10565920?module_item_id=2623346

Best,
George

Derek Och

Hey Ayush and Joseph,

I liked the Avidex idea and how the database connects birds, sightings, nests, rarities, and rewards. The purpose is explained clearly, and the data types generally fit the information being stored.

The main issue I noticed is that BirdsNests, BirdersRewards, and BirdsList do not have primary keys in the SQL. Since each table represents a relationship, both foreign keys should form a composite primary key. This would prevent duplicate relationships and match the normalization section.

There are also some differences between the outline and SQL. The outline says birderName, gpsLocation, time, and threshold are required, but the SQL allows NULL values. BirderName is also described as unique, but the SQL does not include a UNIQUE

constraint. BirdsNest and BirdsNests are also both used, so the naming should be consistent.

BirdsList may create a normalization issue because its totals can be calculated from Sightings. Storing the totals in both places could cause inconsistent data if one table is updated without the other.

I also noticed that Blue Jay uses Unommon instead of Uncommon, and BirdersRewards references rewardID 4 even though only rewards 1 through 3 are inserted. These references should be corrected. CASCADE may also be useful in the relationship tables so related records are removed when a parent record is deleted.

Overall, the project has a creative idea and a good foundation. Good work.

This review was done by me without the use of AI.

Elisha Roche

Hello There, Ayush and Joseph from Group 28, here are my two cents about your projects:

The SQL matches the ERD and outline well. One thing I noted is that gpsLocation and time are listed as NOT NULL in the outline but the SQL allows NULL for both, the same for threshold in Rewards. The outline says NOT NULL but the SQL says NULL. birderName is listed as NOT NULL and UNIQUE in the outline, but in the SQL it's NULL with no UNIQUE constraint. These should be fixed to match the outline.

BirdsNests, BirdersRewards, and BirdsList are all missing composite primary keys and since each of these tables holds pairs of foreign keys, the combination of those two foreign keys should be declared as the primary key because without it, duplicate rows can be inserted. For Example, the same birder could be linked to the same reward more than once, which shouldn't be allowed.

One thing I noticed in the Rewards description column is VARCHAR(45) in the SQL, but based on the sample data ("Spotted your first 10 birds!") a longer length like VARCHAR(255) would make more sense. Also, the photographUrl column is VARCHAR(45) in the SQL, but the sample URLs are already over 30 characters, so that size will likely cause issues when real URLs are used.

Almost all the sample data from the PDF got inserted. One problem I found is BirdersRewards inserts rewardID 4, but only 3 rewards are inserted into the Rewards table.

This would cause a foreign key violation. Also, the reward name in the SQL is Amatuer Birder but the PDF has it as Amateur Birder, that's a typo you should probably fix.

The BirdsList INSERT doesn't name the columns explicitly, the values are just (count, birdID, birderID) which are based on the column order in the CREATE TABLE. I'd suggest naming the columns in the INSERT statement so the code is easier to maintain.

BirdsList stores a running count of how many times each birder has seen a specific bird, but that same count can be calculated from Sightings by summing birdCount grouped by birderID and birdID. Storing derived data in two places means they can get out of sync if one gets updated without the other.

The comments are well written, on the other hand no CASCADE rules are set on any foreign key relationships. For example when deleting a bird, it won't auto clean up rows in Sightings, BirdsNests, or BirdsList that reference it. Adding ON DELETE CASCADE to those relationship tables would prevent orphaned records.

There's also a typo in the Bald Eagle wingspan insert and Red-tailed hawk row which is wingspsan instead of wingspan.

I have done this review 100% on my own.

Yelyzaveta Bulyhina

The schema and ERD match well. There are a couple of typos in the data, like 'propogate', 'knwon', and 'wingspsan', plus 'Amatuer Birder' where the PDF has "Amateur." All tables are plural and attributes singular, with consistent capitalization, and the schema is easy to read and clear. The main issue is that none of the intersection tables have a primary key (BirdsNests, BirdersRewards, and BirdsList). Since each holds a pair of foreign keys, that combination should be the composite primary key. Without it, duplicate rows can be inserted so the same birder could be linked to the same reward more than once. There are also normalization issues. There is redundancy in Sightings, which duplicates BirdsList, and the same applies to Birders.points, since that count can be calculated from Sightings by summing birdCount grouped by birderID and birdID. The SQL file is syntactically correct. However, as other reviewers stated, VARCHAR(255) would be more appropriate than VARCHAR(45) for the Rewards description and photographUrl. A few columns (birderName, gpsLocation, time, threshold) are NOT NULL in the outline but allow NULL in the SQL, and birderName is missing its UNIQUE constraint. Foreign keys are properly defined and match the schema, and all relationships are present, though no CASCADE rules are set. Almost all example data is inserted, but Blue Jay uses 'Unommon' and BirdersRewards references rewardID 4 when only 3 rewards exist, both causing FK violations. Overall the SQL is well structured and commented. Great work!

Actions Based on Feedback Step 2

- Rearranged ERD diagram and Schema diagram so it's more legible
- Added composite primary keys to our three intersection tables
- Fixed various spelling typos
- Added ON DELETE CASCADE rules to the parent entities
- Changed the outline to properly represent Sighting.gpsLocation and Sighting.time to allow NULL values
- Reformatted and fixed BirdersRewards to properly show the 3 Rewards for the 4 Birders
- Changed the DDL.sql so it doesn't require a DB named `Avidex` to run

Title Page

Avidex Project [Working Title]

A Gamified Birding Collection App

A Formal Proposal Prepared for Ms. Visor Super
640A Kerr Administration Building, Corvallis, OR 97331

August 13, 2026
CS340 Section 400

Created By	
Ayush Baruah	Joseph Eidel

Overview

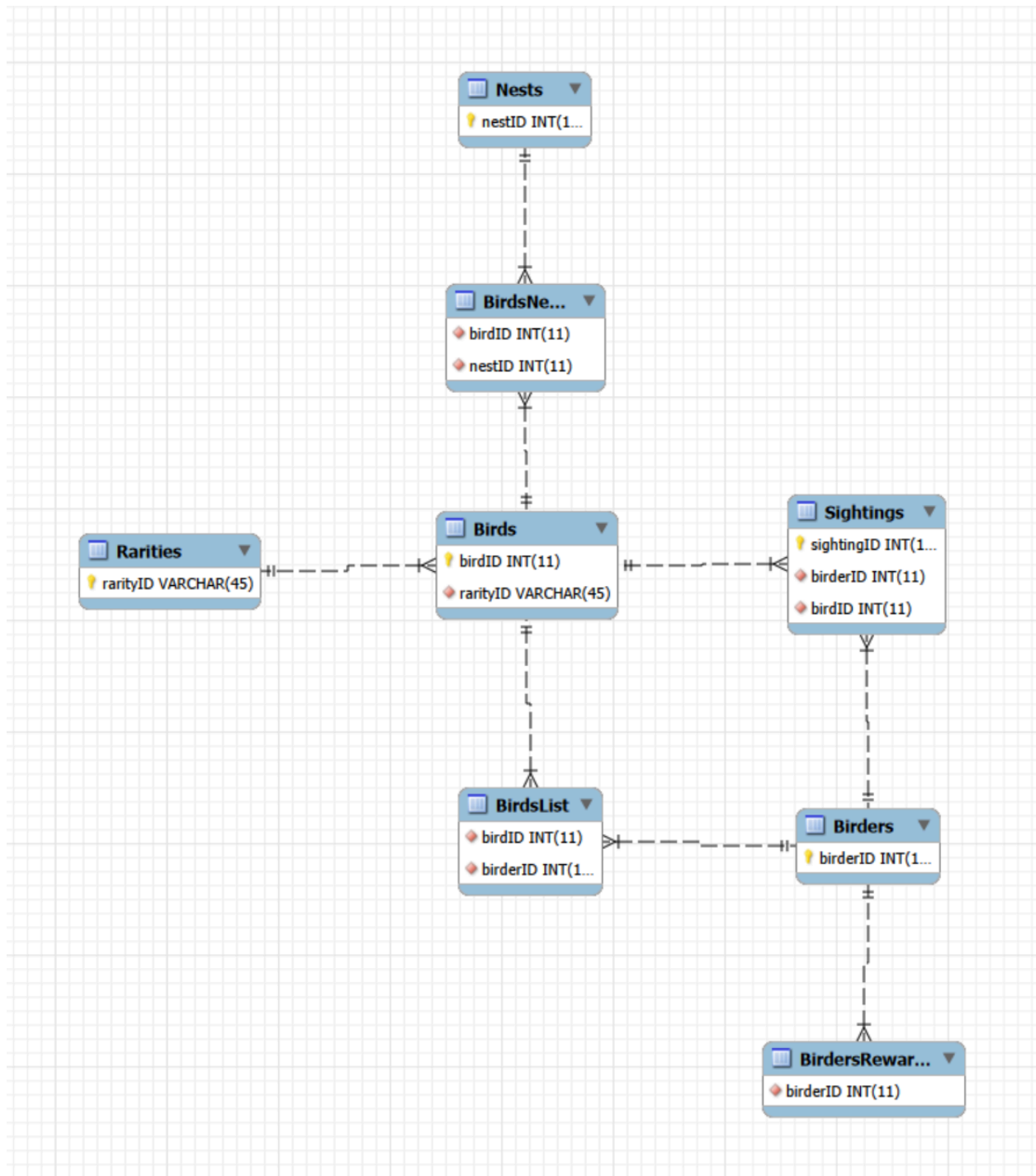
The Avidex Project team requests that Ms. Visor Super review the following Formal Proposal and approve it for further development. Specifically, the Avidex Project team requests that a gamified birding application be developed. Despite the strong interest in birding, we feel there is a severe lack of applications for tracking a birder's sightings in a fun and unique way. Therefore, we expect to create a birdwatching application meant for birding enthusiasts and newcomers. We're catering to a demographic of about 96 million birders in the U.S. — about $\frac{1}{3}$ of the population — and hope to help at least 20,000 birders keep track of 2,000 sightings each! With a Pokédex-like UI and setup, our application will encourage users to go out into the world and look for birds in nature to earn incremental rewards. Soon enough, users will go out into the world and "collect" up to 1,107 birds within their Avidex.

Database Outline

- **Birds-** records the details of Bird species that will populate the Avidex
 - birdID: int, PK, AI, UQ, not NULL — An integer to keep track of the specific bird's ID
 - commonName: varchar, not NULL — The bird species' common name i.e. American Crow, Blue Jay
 - species: varchar, UQ, not NULL — The specific bird's species name i.e. Corvus Corax, Cyanocitta cristata
 - callUrl: varchar, NULL — A URL to link to the call of the bird
 - wingspan: varchar, not NULL — A short description of the range of the wingspan for the bird
 - size: varchar, not NULL — A short description of the size of the bird
 - identifyingMarks: text, not NULL — A description of the identifying features of the bird
 - range: text, not NULL — A description of the bird's range
 - description: text, not NULL — Any interesting information about the bird's lifestyle and characteristics
 - photographUrl: varchar, not NULL — A URL to link to image(s) of the bird
 - matingSeason: varchar, not NULL — A description of the mating season of the bird
 - rarityID: varchar, FK, not NULL — The rarity of the bird
 - Relationships:
 - 1:M relationship to Sightings with a FK of Sightings.birdID
 - 1:M relationship to BirdsNest with a FK of BirdsNests.birdID
 - M:1 relationship to Rarity with a FK of Birds.rarityID
- **Rarities-** categorical table that records the rarity of each Bird
 - rarityID: varchar, PK, UQ, not NULL — The rarity of the bird in our app, i.e., Common, Uncommon, Rare, Legendary
 - Relationships:
 - 1:M relationship to Birds
- **BirdsNests-** composite entity for the M:N relationship of Birds to Nests
 - nestID: int, FK, not NULL — A foreign key for the Nests list
 - birdID: int, FK, not NULL — A foreign key for the Birds list
 - Relationships:
 - M:1 relationship to Nest with a FK of BirdsNests.nestID
 - M:1 relationship to Birds with a FK of BirdsNests.birdID
- **Nests-** records the details of various nests that birds build
 - nestID: int, AI, PK, UQ, not NULL — An ID for this type of nest
 - type: varchar, not NULL — The type of nest this bird has, i.e., Scrape, Cup, Burrow
 - location: varchar, not NULL — The location of this type of nest, i.e., Cliff, Brush, Tree
 - Relationships:
 - 1:M relationship to BirdsNests with a FK of BirdsNests.nestID
- **Sightings-** a transaction table to record the individual bird sightings
 - sightingID: int, PK, AI, UQ, not NULL — The ID for the sighting
 - birderID: int, FK, not NULL — The foreign key to link back to the Birder who made the sighting
 - birdID: int, FK, not NULL — The foreign key for the bird that was sighted
 - birdCount: int — The # of the specific bird they saw
 - gpsLocation: point, NULL — The location they saw it at
 - time: datetime, NULL — The time and day the bird was spotted
 - Relationships:
 - M:1 relationship with Birds with an FK of Sightings.birdID
 - M:1 relationship with Birders with an FK of Sightings.birderID
- **BirdsList-** tracks the list of birds a user has seen and the total count
 - birderID: int, FK — the ID for the users
 - birdID: int, FK — the bird they saw
 - count: int NOT NULL — The number of times they saw the specified bird
 - Relationships
 - 1:M relationship with Birders with FK of BirdsList.birderID

- 1:M relationship with Birds with a FK of BirdsList.birdID
- **Birders-** user entity
 - birderID: int, PK, AI, UQ, NOT NULL — The id of the birder
 - birderName: varchar, UQ, NOT NULL — The name of the birder
 - points: int — The amount of points this birder has
 - Relationships:
 - 1:M relationship with BirdsList with a FK of BirdsList.birderID
 - 1:M relationship with BirdersRewards a FK of BirdersRewards.birderID
- **BirdersRewards-** composite entity for the M:N relationship of Birder to Rewards
 - birderID: int, FK, not NULL — The ID of the birder
 - rewardID: int, FK, not NULL — The ID of the reward
 - Relationships:
 - M:1 relationship to Birders with a FK of BirdersRewards.birderID
 - M:1 relationship to Rewards with a FK of BirdersRewards.rewardID
- **Rewards-** categorical entity for types of rewards
 - rewardID: int, PK, AI, UQ, NOT NULL — The ID of this reward
 - name: varchar, UQ, NOT NULL — The name for the reward
 - description: varchar, NOT NULL — A description of the reward
 - threshold: int, NOT NULL — The minimum amount of points needed to earn the reward
 - Relationships: 1:M with BirdersRewards with a FK BirdersRewards.rewardID

ER Diagram:



Schema Design and Normalization:

We started by writing out a sample sighting data table, with one row per observation and columns for the birder, the bird, its rarity, its nest styles, and the observation details. The report violated first normal form in one place. Because a species may build more than one style of nest, the nest column held a repeating group. The Bald Eagle row listed both a platform nest in a tree and a platform nest on a cliff in a single cell. We removed the repeating group by extracting nests into their own entity and joining it to Birds through the BirdsNests intersection table, which stores one row per bird/nest pairing. Every attribute now holds a single value and every table has a defined primary key, so the design is in 1NF.

For second normal form we looked at our three intersection tables. BirdsNests and BirdersRewards have no non key attributes, making partial dependency impossible. An earlier draft carried a description attribute on BirdsNests so we removed it because it described the nest style rather than the pairing, which made it partially dependent on nestID alone. That information now lives in Nests. BirdsList has one non-key attribute, count, which is meaningless without both halves of its key since it records how many times this birder saw this bird. Every other table uses a single-attribute primary key. The design is therefore in 2NF.

We then looked for transitive dependencies. Rarity was repeated on every row of the report and is now a separate entity keyed on the label itself, with Birds carrying only the foreign key, so each tier is stored once. Nest type and location describe the nest rather than the sighting and are stored once per nest record. Bird attributes such as wingspan, range, and description depend on the bird rather than the sighting event and are stored once in Birds.

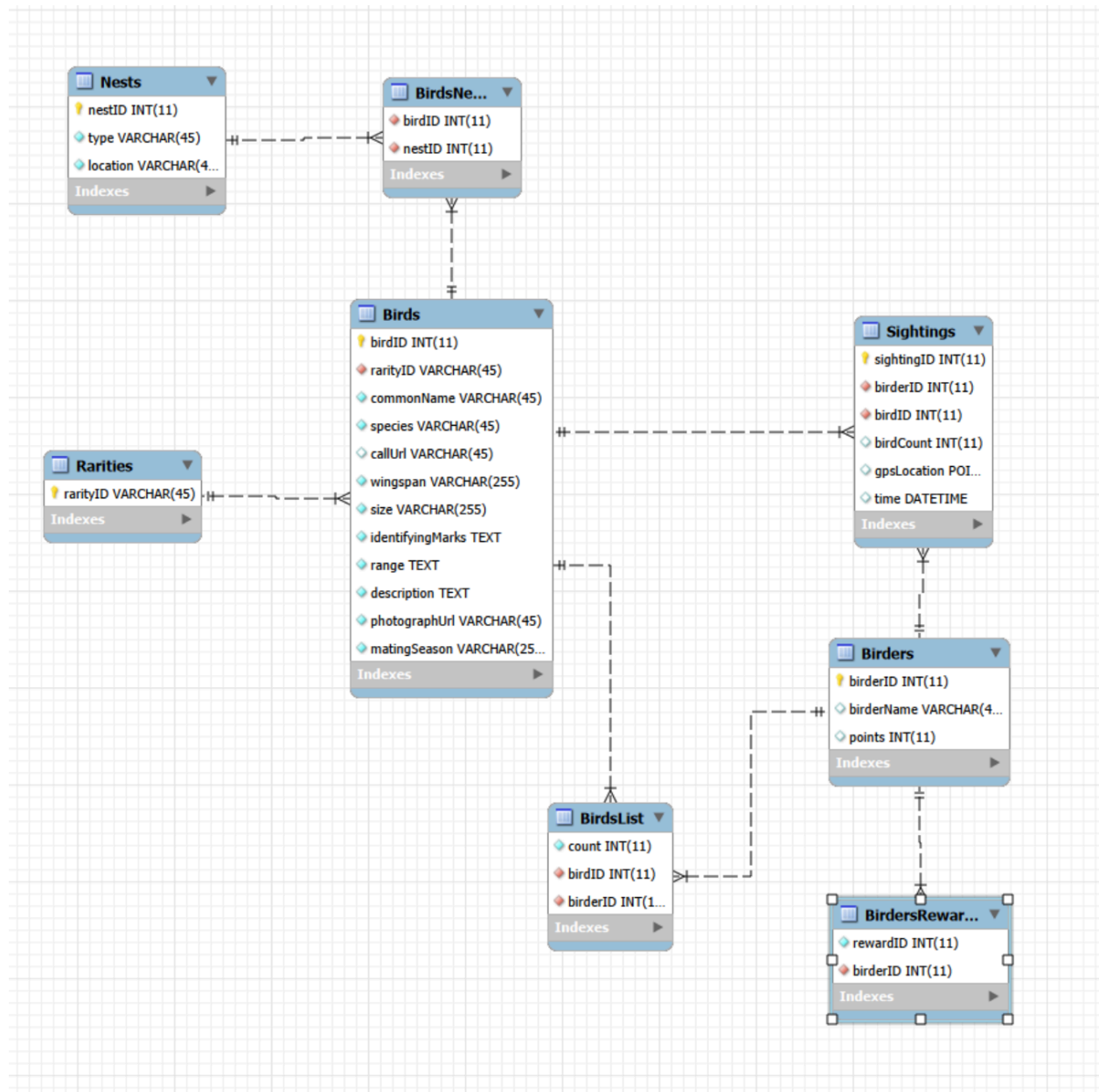


Table: Rarities

rarityID
Common
Uncommon
Rare
Legendary

Table: Birds (part 1 of 3 — identity and media)

birdID	commonName	species	rarityID	callUrl	photographUrl
1	Bald Eagle	Haliaeetus leucocephalus	Uncommon	https://bald eaglecallhere.com	https://bald eaglephotoshere.com
2	American Robin	Turdus migratorius	Common	https://americanrobincallhere.com	https://americanrobinphotoshere.com
3	Blue Jay	Cyanocitta cristata	Uncommon	https://bluejaycallhere.com	https://bluejayphotoshere.com
4	Northern Cardinal	Cardinalis cardinalis	Uncommon	https://northerncardinalcallhere.com	https://northerncardinalphotoshere.com
5	Red-tailed Hawk	Buteo jamaicensis	Common	https://red-tailedhawkcallhere.com	https://red-tailedhawkphotoshere.com
6	Great Blue Heron	Ardea herodias	Uncommon	https://greatblueheroncallhere.com	https://greatblueheronphotoshere.com
7	American Crow	Corvus brachyrhynchos	Common	https://americancrowcallhere.com	https://americancrowphotoshere.com
8	Common Raven	Corvus corax	Common	https://commonravencallhere.com	https://commonravenphotoshere.com
9	Mourning Dove	Zenaidura macroura	Common	https://mourningdovecallhere.com	https://mourningdovephotoshere.com

Table: Birds (part 2 of 3 — measurements and season)

birdID	wingspan	size	matingSeason
1	The average Bald Eagle wingspan is 6.9 feet	The average Bald Eagle is 27.9 inches to 37.8 inches long	Bald Eagle nesting season typically begins in December, and lasts until July. Though their courtship behaviors may begin as early as late Fall, depending on location.
2	The average American Robin wingspan is 12.2 to 15.8 inches	The average American Robin is 7.9 to 11 inches long	American Robin nesting season begins as early as March in warmer regions, with most activity occurring through April until July. Pairs may raise two or three broods per season.
3	The average Blue Jay wingspan is 13.4 to 16.9 inches	The average Blue Jay is 9.8 to 11.8 inches long	Blue Jay nesting season begins mid-March up to July
4	The average Northern Cardinal wingspan is 9.8 to 12.2 inches	The average Northern Cardinal is 8.3 to 9.1 inches long	Northern Cardinal nesting season lasts between March and September
5	The average Red-tailed Hawk wingspan is 44.9 to 52.4 inches	The average female Red-tailed Hawk is 19.7 to 25.6 inches long while the males are 17.7 to 22.1 inches long	Red-tailed Hawk nesting season lasts from February to early mid-March
6	The average Great Blue Heron wingspan is 5.5 to 6.6 feet	The average Great Blue Heron is 3.2 to 4.5 feet long	Great Blue Heron nesting season lasts from April to May
7	The average American Crow wingspan is 2.8 to 3.3 feet	The average American Crow is 1.3 to 1.7 feet long	American Crow nesting season lasts from February to May

birdID	wingspan	size	matingSeason
8	The average Common Raven wingspan is 3.81 to 3.88 feet	The average Common Raven is 1.84 to 2.27 feet long	Common Raven nesting season lasts from February to May
9	The average Mourning Dove wingspan is 1.48 feet	The average Mourning Dove is 9.1 to 13.4 inches long	Mourning Dove nesting season lasts from March to May

Table: Birds (part 3 of 3 — descriptive text)

birdID	identifyingMarks	range	description
1	The Bald Eagle is easily identified by its white capped head and neck, in contrast to the rest of its dark brown coat	Found all across North America, except for the most northern regions and below Mexico	The Bald Eagle got its name from the Middle English word, 'Balde', meaning white-headed (not hairless!) These eagles mainly eat fish, and can be found around bodies of water. Though more often than not, they prefer to steal fish from other fishing animals, humans included.
2	The American Robin can be identified by their dark heads, warm orange underbellies, gray-brown bodies, and white patches under their tails	Found all across North America	American Robins are common all across North America, known for being found year-round anywhere south of Canada. Birds that breed above that dividing line leave for the U.S. when winter approaches. Early signifiers of Spring, they can be seen foraging through lawns during this time.
3	The Blue Jay is white to light grey underneath, and has various shades of blue along its back and crest. Its head is circled with a necklace of black feathers.	Found mainly on the eastern side of the midwestern U.S. and southeastern side of Canada	Blue Jays are intelligent, known to hide away acorns in various locations before winter arrives, and remember most of them during and after winter. The ones they forget help propagate new trees. They have been observed rehidng their acorns if they notice another Blue Jay sees them burying it, preventing the theft of their nutritious treasure.
4	The Northern Cardinal has a reddish beak surrounded by black around its edges. Males are brilliant red all over. Females are pale brown with reddish tinges in their wings, tail, and crest.	They are found mainly on the eastern side of the midwestern U.S. and along the northeast coast of Mexico	Northern Cardinals are very popular birds, being the state bird for seven U.S. states. They also are one of the few songbird species that have females that sing as well. Both males and females can often be seen aggressively attacking their reflections in spring and early summer, when they are the most territorial.
5	The Red-tailed Hawk can be identified by a brown coat above and a pale coat below, with a streaked belly and a dark bar between the shoulder and wrist on the underside of its wings. Its tail is pale below, and a cinnamon-red above.	They are found all across North America, with their range spanning all the way from Canada to the northernmost regions of Nicaragua, and even the Caribbean Islands.	Red-tailed Hawks have screeches that are extremely raspy and shrill. Often when you see eagles and other hawks on television, the call you hear is almost always one of a Red-tailed Hawk. They are also known for swooping courting rituals, and hunting in pairs. They are the most common hawk in North America.

birdID	identifyingMarks	range	description
6	The Great Blue Heron can be identified by its distinct blue-gray colour, and wide black stripe over the eye and head. In flight, its wings are two-toned, pale on the forewing and darker on the flight feathers. A subspecies in coastal southern Florida is known to be pure white.	The Great Blue Heron lives all across the U.S., though they can be seen all across Central America during the nonbreeding season and found in southern midwest Canada and northern midwest U.S.A. during the breeding season.	Great Blue Herons, like most Herons, are fishing birds. They are often found wading through the edges of rivers, ponds, and lakes. They often move slowly while peering into the depths before striking at speed at fish.
7	The American Crow can be identified by its all-black color and distinctive hoarse, cawing call. They tend to have fan-shaped tails, as opposed to the diamond-shaped tails of their cousin, the Raven.	The American Crow lives all across the U.S., save for the southwestern deserts, and can be seen as far north as Canada during their breeding season.	American Crows eat mainly earthworms, insects, small animals, seeds, and fruit. However, they aren't picky and eat garbage, carrion, and chicks from other nests. Known widely for their cunning, they have been known to distract other animals to steal food from them.
8	The Common Raven can be identified by its all-black coat. Often larger than crows, they have diamond-shaped tails, and a deeper, croaking call, compared to Crows.	The Common Raven lives on the eastern side of the U.S. Midwest, and all of the area north of Canada. They also live on the coasts of Greenland, and nearly all of Eurasia.	Ravens are known for their superior intelligence, even compared to other members of the Corvid family. They have been solving ever more complex problems given to them by ever more creative scientists.
9	The Mourning Dove can be identified by its plump body, long tail, and relatively small head. They're often brown to tan overall with black spots on their wings, and black-bordered white tips on their tails.	Mourning Doves live all across the U.S. and Mexico, with some territory even being in the southernmost regions of Canada, and the Caribbean islands.	Mourning Doves feed on seeds on the ground, pecking their way through open country or lawns. On the daily, they eat 12 to 20 percent of their body weight, and can drink water that's up to almost half the salinity of the sea.

Table: Nests

nestID	type	location	nestID	type	location	nestID	type	location
1	Scrape	Tree	21	Domed	Brush	41	Globular	Water
2	Scrape	Cliff	22	Domed	Ground	42	Globular	Man-Made Surface
3	Scrape	Brush	23	Domed	Water	43	Cavity	Tree
4	Scrape	Ground	24	Domed	Man-Made Surface	44	Cavity	Cliff
5	Scrape	Water	25	Pendulous	Tree	45	Cavity	Brush
6	Scrape	Man-Made Surface	26	Pendulous	Cliff	46	Cavity	Ground
7	Platform	Tree	27	Pendulous	Brush	47	Cavity	Water
8	Platform	Cliff	28	Pendulous	Ground	48	Cavity	Man-Made Surface
9	Platform	Brush	29	Pendulous	Water	49	Burrow	Tree
10	Platform	Ground	30	Pendulous	Man-Made Surface	50	Burrow	Cliff

nestl D	type	location	nestl D	type	location	nestl D	type	location
11	Platform	Water	31	Pensile	Tree	51	Burrow	Brush
12	Platform	Man-Made Surface	32	Pensile	Cliff	52	Burrow	Ground
13	Cup	Tree	33	Pensile	Brush	53	Burrow	Water
14	Cup	Cliff	34	Pensile	Ground	54	Burrow	Man-Made Surface
15	Cup	Brush	35	Pensile	Water	55	Mound	Tree
16	Cup	Ground	36	Pensile	Man-Made Surface	56	Mound	Cliff
17	Cup	Water	37	Globular	Tree	57	Mound	Brush
18	Cup	Man-Made Surface	38	Globular	Cliff	58	Mound	Ground
19	Domed	Tree	39	Globular	Brush	59	Mound	Water
20	Domed	Cliff	40	Globular	Ground	60	Mound	Man-Made Surface

Table: BirdsNests

birdID	nestID
1	7
1	8
2	14
3	14
4	15
5	7
5	8
6	7
7	7
8	8
9	7
9	10

Table: Birders

birderID	birderName	points
1	Joseph Eidel	13
2	Ayush Baruah	15
3	Leonel Messi	9
4	David Attenborough	61

Table: Sightings

sightingID	birderID	birdID	birdCount	gpsLocation	time
1	1	7	4	NULL	NULL
2	1	9	2	NULL	NULL
3	1	4	2	NULL	NULL
4	1	7	3	NULL	NULL
5	2	7	5	NULL	NULL
6	2	2	1	NULL	NULL
7	2	5	2	NULL	NULL
8	2	1	1	NULL	NULL
9	2	8	2	NULL	NULL
10	2	6	2	NULL	NULL
11	2	9	3	NULL	NULL
12	3	8	5	NULL	NULL
13	3	3	2	NULL	NULL
14	3	9	2	NULL	NULL
15	4	1	3	NULL	NULL
16	4	2	3	NULL	NULL
17	4	3	7	NULL	NULL
18	4	4	13	NULL	NULL
19	4	7	35	NULL	NULL

Table: BirdsList

birderID	birdID	count
1	7	7
1	9	2
1	4	2
2	7	5
2	2	1
2	5	2
2	1	1
2	8	2
2	6	2
2	9	3
3	8	5
3	3	2
3	9	2

birderID	birdID	count
4	1	3
4	2	3
4	3	7
4	4	13
4	7	35

Table: Rewards

rewardID	name	description	threshold
1	On My Way	Spotted your first 10 birds!	10
2	Amateur Birder	Spotted 25 birds!	25
3	True Birder	Spotted 50 birds!	50

Table: BirdersRewards

rewardID	birderID
1	1
1	2
1	4
2	4
3	4

Citations

All About Birds, Cornell Lab, www.allaboutbirds.org/guide/. Accessed 23 July 2026.

Audubon, John James. "Bird Nesting Habits by Species: Fascinating Insights Revealed - Birds." Birdopedia, 13 Nov. 2025, birdopedia.com/bird-nesting-habits-by-species/.

"Bald Eagle Physical Characteristics | American Eagle Foundation." American Eagle Foundation, eagles.org/what-we-do/educate/learn-about-eagles/bald-eagle-physical-characteristics/. Accessed 23 July 2026.

Carver, Erin. U.S. Department of the Interior, U.S. Fish and Wildlife Service, 2024, Birding in the United States: A Demographic and Economic Analysis, <https://www.fws.gov/sites/default/files/documents/2024-11/2022-birding-in-the-us-demographic-and-economic-analysis.pdf>. Accessed 7 July 2026.

Chesser, R. T., et al. "Check-List of North American Birds (Online)." American Ornithological Society, 2025, checklist.americanornithology.org/taxa/.

Thompson, Mya, and Marla Coppelino. "How Nest Designs Protect Eggs and Baby Birds | Bird Academy • the Cornell Lab." Cornell Lab • Bird Academy, Cornell, 6 Mar. 2026, academy.allaboutbirds.org/understand-how-nest-designs-protect-baby-birds/.

Wild-Bird-Watching, www.wild-bird-watching.com/. Accessed 23 July 2026.